Light

Cyber Security — Prevention

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OCR GCSE Computer Science — **Identifying and Preventing Vulnerabilities (J277/01 Section 1.4)**. This topic covers methods used to protect computer systems and networks from cyber attacks, including penetration testing, network policies, firewalls, encryption, anti-malware software, user access levels, password policies, and physical security measures.

Exam questions typically ask you to **describe** prevention methods, **explain** how they protect against specific threats, and **recommend** appropriate security measures for given scenarios. Questions range from 1-mark definitions to 6-mark discussions requiring you to evaluate multiple security measures.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and give an example where appropriate.

- **Describe** penetration testing and explain why organisations use ethical hackers.
- **Explain** the purpose of network forensics in investigating security incidents.
- **Describe** network policies and explain how they help maintain security.
- **Explain** how anti-malware software detects and removes malicious software.
- **Describe** what a firewall does and how it protects a network.

- **Explain** user access levels and why different users have different permissions.
- **Describe** what makes a strong password and explain password policies.
- **Explain** how encryption protects data and the difference between encrypting data at rest and in transit.
- **Describe** physical security measures and explain why they are important.

> **Specification note.** OCR J277 Section 1.4 requires understanding of methods to identify vulnerabilities (penetration testing, network forensics) and methods to prevent attacks (network policies, anti-malware, firewalls, user access levels, passwords, encryption, physical security). Questions often link prevention methods to specific attack types from the threats section.

Toolkit

Prevention methods — overview

Method	Purpose	Protects against
Penetration testing	Find vulnerabilities before attackers do	All attack types (proactive identification)
Network forensics	Investigate and learn from security incidents	Future attacks (by understanding past ones)
Network policies	Define acceptable use and security procedures	Insider threats, social engineering
Anti-malware	Detect and remove malicious software	Viruses, worms, trojans, spyware, ransomware
Firewalls	Filter network traffic based on rules	Unauthorised access, some malware, DoS attacks
User access levels	Limit what users can access and modify	Insider threats, malware spread, accidental damage
Passwords	Authenticate users before granting access	Unauthorised access, brute force attacks
Encryption	Make data unreadable without the key	Data interception, theft, unauthorised access
Physical security	Protect hardware from physical access	Theft, tampering, unauthorised physical access

Defence in depth

Defence in depth is the principle of using multiple layers of security. No single security measure is perfect, so organisations combine several methods to create

overlapping protection. If one layer fails, others remain to protect the system. For example, a company might use: a firewall (to filter traffic) + anti-malware (to catch threats that get through) + user access levels (to limit damage if a user is compromised) + encryption (to protect data even if stolen) + physical security (to prevent hardware theft).

Strong password criteria

A strong password should include:

- At least 8-12 characters (longer is better)
- A mix of uppercase and lowercase letters
- Numbers (0-9)
- Special characters (!@#$%^&*)
- No dictionary words or personal information
- Not reused across multiple accounts

Why this matters: Each additional character type exponentially increases the number of possible combinations, making brute force attacks much slower. A password using only lowercase letters has 26 options per character; adding uppercase, numbers, and symbols increases this to over 90 options per character.

> **Exam tip:** When asked about preventing a specific attack, name the prevention method and explain **how** it stops that particular attack. Generic answers like "use security software" will not score full marks.

1. Testing and policies

Before implementing technical defences, organisations need to understand their vulnerabilities and establish clear rules for how systems should be used. Penetration testing identifies weaknesses, network forensics investigates incidents, and network policies set the rules that users must follow.

1.1 Penetration testing

Penetration testing (also called "pen testing" or "ethical hacking") is the practice of deliberately attempting to break into a system to find vulnerabilities before malicious hackers do. Organisations hire professional security experts — **ethical hackers** — to attack their systems in a controlled way.

How penetration testing works:

1. **Planning:** Define the scope — which systems will be tested and what methods are allowed
2. **Reconnaissance:** Gather information about the target systems
3. **Scanning:** Use automated tools to identify potential vulnerabilities
4. **Exploitation:** Attempt to exploit discovered vulnerabilities
5. **Reporting:** Document all findings and recommend fixes
6. **Remediation:** The organisation fixes the identified vulnerabilities

Why use penetration testing:

- Identifies vulnerabilities before criminals find them
- Tests whether existing security measures are effective

- Helps prioritise security investments
- May be required for compliance with regulations
- Provides evidence of due diligence in security

Penetration testers must have written permission before testing. Without permission, the same activities would be illegal hacking.

1.2 Network forensics

Network forensics is the process of capturing, recording, and analysing network traffic and events to investigate security incidents. When a breach occurs, forensic analysis helps determine what happened, how it happened, and what data was affected.

Uses of network forensics:

- **Incident investigation:** Determine how an attack occurred and what was compromised
- **Evidence gathering:** Collect evidence for legal proceedings or insurance claims
- **Attack attribution:** Try to identify who was responsible for an attack
- **Improving defences:** Learn from incidents to prevent similar attacks
- **Compliance:** Demonstrate that incidents were properly investigated

Network forensics relies on **logs** — records of events that occur on systems and networks. Organisations must ensure logging is enabled and logs are stored securely and retained for an appropriate period.

1.3 Network policies

Network policies are formal documents that define rules and procedures for using an organisation's network and computer systems. They tell users what they can and cannot do, and what security practices they must follow.

Common types of network policies:

- **Acceptable Use Policy (AUP):** Defines what users are allowed to do on the network (e.g., no personal file sharing, no accessing inappropriate content)
- **Password policy:** Requirements for password strength, how often passwords must be changed, rules against sharing passwords
- **Email policy:** Rules about email attachments, recognising phishing, what can be sent externally
- **BYOD policy:** Rules for "Bring Your Own Device" — personal phones/laptops connecting to the network
- **Data handling policy:** How sensitive data should be stored, transmitted, and disposed of
- **Incident response policy:** What to do if a security incident is suspected

Why network policies matter:

- Set clear expectations for user behaviour
- Reduce risk of accidental security breaches
- Provide grounds for disciplinary action if rules are broken
- Help defend against social engineering (users know to be suspicious)
- Required for compliance with data protection laws

Policies are only effective if users are trained on them and if they are enforced. A policy that exists but is ignored provides no protection.

> **Human element:** Many security breaches result from human error or social engineering rather than technical exploits. Network policies address the human element of security by educating users and setting behavioural expectations.

Exam tip: Penetration testing is **proactive** (done before an attack to find weaknesses). Network forensics is **reactive** (done after an incident to investigate what happened). Don't confuse these two.

Worked Example 1 — Penetration testing definition

Question: Describe what is meant by penetration testing. [2 marks]

Model answer:
Penetration testing is the practice of deliberately attempting to hack into a system **(1)** to identify vulnerabilities before malicious attackers can exploit them **(1)**.

Exam tip: Include both what it is (authorised hacking attempt) and why it's done (find vulnerabilities first).

Worked Example 2 — Why use ethical hackers

Question: Explain why a company might hire ethical hackers to test their systems. [2 marks]

Model answer:
Ethical hackers can identify security vulnerabilities before criminal hackers find and exploit them **(1)**. This allows the company to fix weaknesses and strengthen their defences proactively **(1)**.

Exam tip: Emphasise the "find it first" benefit — better to have a friendly expert find the hole than a criminal.

Worked Example 3 — Network forensics purpose

Question: State two reasons why an organisation would use network forensics after a cyber attack. [2 marks]

Model answer:
1. To determine how the attack occurred and what vulnerabilities were exploited **(1)**.
2. To gather evidence that could be used in legal proceedings against the attacker **(1)**.

Exam tip: Other valid reasons: identify what data was compromised, learn how to prevent similar attacks, comply with legal requirements to investigate breaches.

Worked Example 4 — Network policies benefit

Question: Explain how having a password policy helps protect a network. [2 marks]

Model answer:
A password policy ensures users create strong passwords that are difficult to guess or crack through brute force **(1)**. It also prevents bad practices like sharing passwords or reusing the same password across multiple accounts **(1)**.

Exam tip: Link the policy to the threat it protects against — password policies defend against brute force attacks and credential theft.

2. Software defences

Software-based security measures form a critical layer of defence. Anti-malware software detects and removes malicious programs, while firewalls control what network traffic is allowed in and out. Encryption protects data even if other defences fail.

2.1 Anti-malware software

Anti-malware software (often called antivirus software) is designed to detect, prevent, and remove malicious software including viruses, worms, trojans, spyware, ransomware, and other threats.

How anti-malware works:

- **Signature-based detection:** Compares files against a database of known malware "signatures" (unique patterns). Effective against known threats but cannot detect new, unknown malware.
- **Heuristic analysis:** Analyses the behaviour and characteristics of programs to identify suspicious activity that might indicate malware. Can detect new threats but may produce false positives.
- **Real-time scanning:** Monitors files as they are opened, downloaded, or executed, blocking threats before they can cause harm.
- **Scheduled scans:** Periodically scans all files on the system to find threats that may have been missed.
- **Quarantine:** Isolates suspicious files so they cannot harm the system while allowing investigation.

Keeping anti-malware effective:

- Regular updates are essential — new malware is created constantly
- Signature databases must be updated daily or more frequently
- The software itself needs updates to detect new attack techniques
- Real-time protection should always be enabled

Anti-malware is not perfect. New "zero-day" malware (malware that exploits previously unknown vulnerabilities) may not be detected until signatures are created. This is why defence in depth is important — anti-malware is one layer, not the only layer.

2.2 Firewalls

A **firewall** monitors and controls incoming and outgoing network traffic based on predetermined security rules. It acts as a barrier between a trusted internal network and untrusted external networks (like the internet).

How firewalls work:

- **Packet filtering:** Examines each data packet and allows or blocks it based on source/destination IP addresses, ports, and protocols
- **Stateful inspection:** Tracks the state of network connections and only allows packets that are part of established, legitimate connections
- **Application-layer filtering:** Can inspect the actual content of traffic to block specific applications or content types

What firewalls protect against:

- Unauthorised access attempts from outside the network
- Some types of malware that try to communicate with external servers
- Data exfiltration (data being sent out without permission)
- Some denial of service attacks

Firewalls can be **hardware-based** (a dedicated device that sits between the network and internet) or **software-based** (a program running on individual computers). Most organisations use both — a network firewall at the perimeter and software firewalls on individual devices.

2.3 Encryption

Encryption converts readable data (plaintext) into an unreadable format (ciphertext) using a mathematical algorithm and a key. Only someone with the correct decryption key can convert the data back to its original form.

Two contexts for encryption:

- **Data at rest:** Encrypting stored data (files on a hard drive, database contents, backup tapes). Protects data if storage media is stolen or accessed without authorisation.
- **Data in transit:** Encrypting data as it travels across networks (HTTPS for web traffic, encrypted email, VPNs). Protects against interception and eavesdropping.

Why encryption is essential:

- Even if other security measures fail, encrypted data remains protected
- Stolen encrypted data is useless without the decryption key
- Intercepted network traffic cannot be read if encrypted
- Required by data protection laws for sensitive personal data
- Protects confidential business information from competitors

Common encryption applications:

- **HTTPS:** Encrypts web traffic between browser and server
- **Full disk encryption:** Encrypts entire hard drives (e.g., BitLocker on Windows)
- **File encryption:** Encrypts individual files or folders
- **Email encryption:** Protects email contents from interception
- **VPN:** Creates an encrypted tunnel for all network traffic
- **Encrypted messaging:** Apps like WhatsApp use end-to-end encryption

Key management: Encryption is only as secure as the protection of the encryption keys. If keys are lost, the data cannot be recovered. If keys are stolen, the encryption provides no protection. Secure key management is critical.

Exam tip: When asked about encryption, specify whether you mean data at rest (stored) or data in transit (being transmitted). This shows understanding of how encryption is applied in different contexts.

Worked Example 5 — Anti-malware function

Question: Describe how anti-malware software protects a computer. [2 marks]

Model answer:
Anti-malware software scans files and compares them against a database of known malware signatures **(1)**. If a match is found, the software blocks or removes the malicious file before it can harm the system **(1)**.

Exam tip: Mention the detection method (signature matching or heuristic analysis) and the action taken (block, quarantine, or delete).

Worked Example 6 — Why update anti-malware

Question: Explain why it is important to keep anti-malware software updated. [2 marks]

Model answer:
New malware is constantly being created by attackers **(1)**. Updates add new malware signatures to the database, allowing the software to detect and protect against the latest threats **(1)**.

Exam tip: Link updates to the constantly evolving threat landscape. Without updates, the software cannot detect new malware.

Worked Example 7 — Firewall purpose

Question: Describe what a firewall does and how it protects a network. [3 marks]

Model answer:
A firewall monitors and filters network traffic based on security rules **(1)**. It examines data packets and blocks those that don't meet the rules (e.g., from

untrusted sources or using blocked ports) **(1)**. This prevents unauthorised access to the network from external attackers **(1)**.

Exam tip: A firewall is a filter/barrier — it controls what goes in and out. Don't confuse it with anti-malware (which specifically targets malicious software).

Worked Example 8 – Encryption for data protection

Question: A company stores customer credit card details. Explain how encryption protects this data. [2 marks]

Model answer:
Encryption converts the credit card data into an unreadable format **(1)**. Even if attackers gain access to the database, they cannot read the data without the decryption key **(1)**.

Exam tip: Emphasise that encrypted data is useless to attackers without the key. This is the core benefit of encryption.

3. Access control and physical security

Controlling who can access what — both logically (user permissions) and physically (building access) — is fundamental to security. Even the best technical defences are useless if anyone can walk in and access the servers directly.

3.1 User access levels

User access levels (also called access control or permissions) determine what resources each user can access and what actions they can perform. The principle is that users should only have access to the resources they need to do their job — this is called the **principle of least privilege**.

Common access levels:

- **Administrator:** Full access to all systems and settings; can install software, change configurations, and manage other users
- **Standard user:** Can use software and access their own files but cannot install software or change system settings
- **Guest:** Very limited access, typically cannot save files or change any settings
- **Custom roles:** Organisations often create specific roles (e.g., "Finance team" with access to financial systems only)

Why user access levels are important:

- **Limits damage from compromised accounts:** If an attacker gains access to a standard user account, they cannot access administrator functions
- **Prevents accidental damage:** Users cannot accidentally delete critical system files they don't have access to

- **Protects sensitive data:** Confidential information is only accessible to authorised personnel
- **Reduces malware impact:** Malware running under a limited user account cannot install system-wide
- **Audit trail:** Knowing who has access helps track who might have accessed or modified data

Access levels should be reviewed regularly. When employees change roles or leave the organisation, their access should be updated or revoked promptly.

3.2 Passwords and authentication

Passwords are the most common form of authentication — proving that you are who you claim to be. A strong password policy is essential for security.

Characteristics of strong passwords:

- Minimum length of 8-12 characters (longer is better)
- Mix of uppercase letters, lowercase letters, numbers, and symbols
- Not based on dictionary words or personal information
- Unique for each account (not reused)
- Not written down or shared

Password policy elements:

- Minimum complexity requirements enforced by the system
- Regular password changes (e.g., every 90 days)
- Account lockout after multiple failed attempts (prevents brute force)
- Password history (prevents reusing recent passwords)
- Prohibition on sharing passwords

Multi-factor authentication (MFA): Using two or more different types of authentication factors provides much stronger security than passwords alone. The three factors are: something you know (password), something you have (phone, token), and something you are (fingerprint, face recognition).

3.3 Physical security

Physical security protects computer systems and data from physical threats — theft, damage, or unauthorised physical access. Technical security measures are useless if an attacker can simply walk in and steal a server.

Physical security measures:

- **Locked doors and access control:** Server rooms and data centres should have locked doors, with access limited to authorised personnel. This may include key cards, PIN codes, or biometric scanners.
- **CCTV:** Security cameras deter criminals and provide evidence if incidents occur.
- **Security guards:** Human guards can respond to incidents and verify identity of visitors.
- **Visitor policies:** Visitors should sign in, wear badges, and be escorted in sensitive areas.
- **Cable locks:** Laptops and other portable equipment can be physically locked to desks.
- **Secure disposal:** Old hard drives and documents must be securely destroyed (shredding, degaussing) to prevent data recovery.
- **Environmental controls:** Fire suppression systems, climate control, and flood protection for data centres.

Why physical security matters:

- Physical access often means complete access — someone at the server can bypass many security controls

- Stolen hardware may contain sensitive data
- Physical damage (fire, flood) can destroy data and systems
- Social engineering often involves physical access (e.g., "tailgating" through secure doors)

> **Tailgating:** A common physical security threat where an unauthorised person follows an authorised person through a secure door. Training users to challenge unknown people and not hold doors for strangers is important.

3.4 Combining defences

Effective security requires combining multiple methods. A layered approach means that if one defence fails, others remain:

- **Perimeter:** Firewalls, physical building security
- **Network:** Network policies, traffic monitoring
- **Endpoint:** Anti-malware on each device, software updates
- **Application:** Secure coding practices, input validation
- **Data:** Encryption, access controls
- **User:** Training, strong passwords, awareness

> **Exam tip:** In scenario questions, recommend multiple security measures. Examiners reward answers that show understanding of defence in depth — no single measure is enough.

Worked Example 9 — User access levels purpose

Question: Explain why organisations use different user access levels. [3 marks]

Model answer:
Different access levels ensure users can only access resources they need for their job (principle of least privilege) **(1)**. If an account is compromised, the attacker only gains access to that user's limited permissions **(1)**. It also prevents accidental damage, as users cannot modify files or settings they don't have permission to access **(1)**.

> **Exam tip:** Link access levels to both malicious threats (limiting attacker access) and accidental problems (preventing user mistakes).

Worked Example 10 — Strong password requirements

Question: State three characteristics of a strong password. [3 marks]

Model answer:
1. Contains a mix of uppercase and lowercase letters **(1)**.
2. Includes numbers and special characters **(1)**.
3. Is at least 8-12 characters long **(1)**.

Exam tip: Other valid characteristics: not a dictionary word, not based on personal information, not reused across accounts, not shared with others.

Worked Example 11 — Physical security measures

Question: A company is setting up a new server room. Describe two physical security measures they should implement. [2 marks]

Model answer:
1. Install locked doors with key card or biometric access control so only authorised IT staff can enter **(1)**.
2. Install CCTV cameras to monitor the room and provide evidence if any security incidents occur **(1)**.

Exam tip: Physical security is often overlooked. Remember: locked doors, CCTV, visitor policies, cable locks, secure disposal.

Worked Example 12 — Defence in depth (6-mark style)

Question: A school stores sensitive student data including medical records and exam results. Discuss the security measures the school should implement to protect this data. [6 marks]

Plan:
- Technical measures (software-based)
- Access control measures
- Physical and policy measures

Model answer:
Technical defences: The school should install a firewall to prevent unauthorised external access to the network. Anti-malware software should be installed on all computers and kept updated to protect against viruses and ransomware. The sensitive data should be encrypted both when stored (at rest) and when transmitted across the network (in transit), so that even if data is stolen, it cannot be read without the encryption key.
Access control: User access levels should ensure only authorised staff (such as the school nurse for medical records, or exam officers for exam results) can access sensitive data. Strong password policies should be enforced, requiring complex passwords and regular changes. Multi-factor authentication could be used for accessing the most sensitive systems.
Physical and policy measures: Servers storing the data should be in a locked room with restricted access. Staff should receive training on data protection,

recognising phishing attempts, and the importance of not sharing passwords. A clear data handling policy should define how sensitive data can be used, who can access it, and how it should be disposed of when no longer needed.

Exam tip: For extended questions about protecting sensitive data, demonstrate defence in depth by covering multiple layers: technical, access control, physical, and human/policy. Link measures to the specific data being protected.

Worked examples summary

The 12 worked examples above cover the main question types you will encounter on cyber security prevention. Here is a summary of the key question patterns:

Example	Question type	Key skill
1	Describe (2 marks)	Penetration testing — authorised hacking to find vulnerabilities
2	Explain why (2 marks)	Ethical hackers find problems before criminals do
3	State reasons (2 marks)	Network forensics — investigate and gather evidence
4	Explain how (2 marks)	Password policy enforces strong passwords
5	Describe how (2 marks)	Anti-malware — signature matching and removal
6	Explain importance (2 marks)	Updates add new malware signatures
7	Describe function (3 marks)	Firewall filters traffic based on rules
8	Explain protection (2 marks)	Encryption makes stolen data unreadable
9	Explain why (3 marks)	Access levels limit damage from compromise
10	State characteristics (3 marks)	Strong password criteria
11	Describe measures (2 marks)	Physical security for server room
12	Discuss (6 marks)	Defence in depth with multiple layers

Quick reference

Key terms — exam-ready definitions

Term	Definition (mark scheme style)
Penetration testing	Authorised attempt to hack into a system to identify vulnerabilities before malicious attackers can exploit them
Ethical hacker	A security professional hired to perform penetration testing with permission
Network forensics	Capturing and analysing network traffic and events to investigate security incidents
Network policy	Formal document defining rules and procedures for using an organisation's network and systems
Anti-malware	Software that detects, prevents, and removes malicious software from a computer
Firewall	Security system that monitors and controls network traffic based on predetermined rules
User access levels	Different permission levels that determine what resources users can access and modify
Encryption	Converting data into an unreadable format using a key; only those with the key can decrypt it
Data at rest	Data that is stored (on a hard drive, in a database, etc.)
Data in transit	Data that is being transmitted across a network
Physical security	Measures to protect hardware and data from physical threats like theft, damage, or unauthorised access
Defence in depth	Using multiple layers of security so that if one fails, others remain to protect the system

Prevention method vs threat

Threat	Prevention methods
Malware	Anti-malware, firewalls, user access levels, user training
Brute force	Strong passwords, account lockout, multi-factor authentication
Social engineering	User training, network policies, verification procedures
Data interception	Encryption (especially for data in transit), VPNs

Threat	Prevention methods
Unauthorised access	Firewalls, user access levels, strong passwords, physical security
Theft (physical)	Physical security, encryption of stored data, cable locks
SQL injection	Input validation, parameterised queries (covered in programming)
DoS attacks	Firewalls, traffic filtering, bandwidth management

Strong password checklist

✓	Requirement
☐	At least 8-12 characters long
☐	Contains uppercase letters (A-Z)
☐	Contains lowercase letters (a-z)
☐	Contains numbers (0-9)
☐	Contains special characters (!@#$%^&*)
☐	Not a dictionary word
☐	Not based on personal information
☐	Unique (not reused from other accounts)

Typical pitfalls

Pitfall 1 — Confusing penetration testing with network forensics.
Wrong: "Penetration testing investigates attacks after they happen."
Right: Penetration testing is **proactive** (before attacks). Network forensics is **reactive** (after incidents).

Pitfall 2 — Confusing firewall with anti-malware.
Wrong: "A firewall removes viruses from your computer."
Right: A firewall **filters network traffic**. Anti-malware **detects and removes malicious software**. They are different tools.

Pitfall 3 — Vague answers about encryption.
Wrong: "Encryption keeps data safe."

Right: "Encryption converts data into an unreadable format. **Without the decryption key**, the data is useless to attackers."

Pitfall 4 — Forgetting to mention the key in encryption answers.
Wrong: "Encryption scrambles data so hackers can't read it."
Right: Encryption converts data to ciphertext. **Only those with the decryption key** can convert it back.

Pitfall 5 — Not linking prevention to specific threats.
Wrong: "Use a firewall for security."
Right: "Use a firewall to **block unauthorised access attempts** from external attackers trying to connect to the network."

Pitfall 6 — Thinking strong passwords alone are enough.
Wrong: "Strong passwords will prevent all unauthorised access."
Right: Strong passwords are one layer. **Defence in depth** requires multiple measures including user access levels, encryption, and training.

Pitfall 7 — Forgetting physical security.
Wrong: "Technical measures like firewalls and encryption are all that's needed."
Right: Physical security is essential. An attacker with physical access to a server can often bypass technical controls.

Pitfall 8 — Not explaining why anti-malware needs updates.
Wrong: "Anti-malware should be updated regularly."
Right: "Updates add **new malware signatures** to detect recently created threats that weren't in the original database."

Pitfall 9 — Confusing user access levels with authentication.
Wrong: "User access levels check your password."
Right: Authentication (passwords) verifies who you are. **Access levels** determine what you're allowed to do once authenticated.

Pitfall 10 — One-dimensional security recommendations.
Wrong: Suggesting only one security measure.
Right: For scenario questions, recommend **multiple complementary measures** (defence in depth) covering different types of threats.

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